Spacecraft Telemetry Health Assessment — Technical Report

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1. Introduction

During spacecraft operations, flight controllers must analyze downlinked housekeeping telemetry rapidly and coordinate recovery actions during brief ground station contact windows, which typically last between 8 and 12 minutes per orbit. While onboard safety systems respond in milliseconds to isolate immediate failures, they lack historical lookback capabilities and operator log context. Therefore, human-in-the-loop ground monitoring is essential to diagnose subtle anomalies and long-term parameter drifts. Fully autonomous classification is unsuitable because safety-critical missions demand human confirmation, verifiable explainability, and compliance audits before executing commands. To address these operational needs, this project develops a hybrid decision-support system that combines deterministic rule checking with language-model-assisted log interpretation, delivering transparent health alerts that operators can directly verify, audit, and override.

At the design level, machine learning approaches were intentionally avoided for severity classification because spacecraft health monitoring requires deterministic and auditable decision-making. Unlike machine learning models, rule-based systems provide explicit thresholds that can be traced to engineering constraints. This ensures predictability, reproducibility, and safety in failure scenarios. Instead, the artificial intelligence model is restricted to natural language interpretation and explanation generation, where probabilistic behavior does not affect system safety.

2. Dataset and methodology

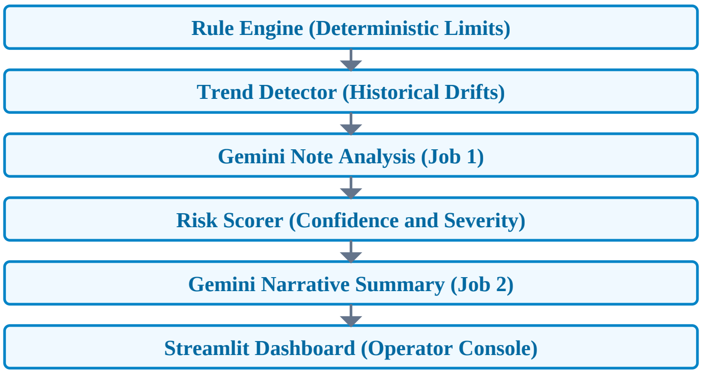
Publicly available spacecraft anomaly benchmarks are designed for training unsupervised machine learning algorithms and are unsuitable for ground station decision support. These datasets lack operator notes, multi-channel correlations, and the structured workflow representation necessary to evaluate diagnostic panels. To build a realistic simulation, we procedurally generated a dataset containing 500 ground station passes across 50 low-Earth orbit satellites. Each satellite has 10 consecutive passes spaced 90 minutes apart to represent sequential orbits.

The telemetry parameters were chosen to cover 5 key satellite subsystems. We used the CubeSat Design Specification and literature only to establish subsystem behavior and physical constraints, not for exact numeric thresholds. The numeric ranges are engineering-derived representative operating envelopes chosen to simulate realistic spacecraft behavior based on literature ranges, rather than mission-specific calibration. The procedural generator simulates correlated variables based on LEO physical constraints rather than random independent fields. For example, during an eclipse pass, solar generation current drops to near zero, solar panels cool, and battery voltage declines sequentially. This physical consistency helps reduce unrealistic parameter combinations that could otherwise produce misleading alerts, helping to model realistic satellite behaviors.

Parameter	Nominal Range	Yellow (caution)	Red (critical)
Battery voltage	24 to 30 V	22 to 24 V	less than 22 V
Battery state of charge	greater than 35%	20% to 35%	less than 20%
Battery temperature	5 to 40 degrees Celsius	0 to 5 or 40 to 45 degrees Celsius	less than 0 or greater than 45 degrees Celsius
Attitude error	less than 3 degrees	3 to 5 degrees	greater than 5 degrees
Bit error rate	less than 1e-6	1e-6 to 1e-4	greater than 1e-4
Processor and memory	less than 80%	80% to 90%	greater than 90%

3. System architecture

The health assessment pipeline is structured as a decoupled, sequential process that isolates deterministic checks from natural language tasks. This separation prevents language model hallucinations from affecting safety-critical limits. The sequential workflow is illustrated in the diagram below, leading from the raw input parameters to the operator dashboard.



The pipeline components coordinate as follows: The rule engine evaluates instantaneous limits and establishes an immutable severity floor. The trend detector runs in parallel, identifying monotonic drifts over the prior 3 passes. A window of 3 passes was selected as the minimal span to detect monotonic drift while preserving responsiveness in short orbital visibility cycles. Once rule and trend flags are compiled, the language model parses the operator note to extract concerns and flag conflicts. The language model serves exclusively in a supporting role to annotate and explain deterministic outputs, ensuring it plays no part in the severity classification decision making. The risk scorer aggregates these outputs to compute the final severity and confidence. The confidence score is an explainability metric indicating uncertainty sources in the rule-based pipeline rather than a statistical probability, starting at 100 percent and reducing via deterministic penalties. Finally, the language model generates the prose reasoning summary, and the Streamlit dashboard renders the complete assessment result, allowing operators to submit annotations without overwriting the calculated severity.

4. System verification and scenario-based evaluation

The system was evaluated using a structured scenario-based verification approach to confirm correct behavior under predefined operational conditions. The five handcrafted scenarios serve as regression tests covering representative operational conditions to verify that deterministic rules, safety floors, trend equations, and confidence penalties execute consistently with design definitions. All five scenarios successfully met their expected results. The system outputs were checked for consistency with the rule definitions, trend logic, and confidence heuristic design across the remaining 450 procedurally generated passes. The system processed these arbitrary, randomized orbital timelines without unexpected rule violations or pipeline crashes during test scenarios.

Scenario	Key Condition	Expected Severity	Result
Nominal pass	All parameters within limits	Nominal, high confidence	Pass
Eclipse battery stress	Voltage in caution range, zero solar current	Monitor or Warning	Pass
Hard limit breach	Voltage below critical red threshold	Critical (rule engine floor)	Pass
Genuine uncertainty	Multi-system yellows, vague operator note	Monitor, low confidence	Pass
Conflicting note	Nominal telemetry, alarming operator note	Nominal, confidence penalty	Pass

- The primacy of hard limits was demonstrated because the critical status in the 3rd scenario was successfully enforced by the rule engine regardless of the note tone.
- The fallback mode was demonstrated because disabling the API key correctly triggered a 15 percent confidence penalty, defaulted the note relationship, and displayed the unavailability banner.
- The operator override mechanism was demonstrated because downgrading the critical pass saved the operator’s decision to a local database while leaving the calculated severity unchanged.

5. Conclusion and limitations

This project successfully demonstrates an explainable decision-support dashboard for satellite operations. By decoupling deterministic mathematical scoring from language model reasoning, the system provides consistent pass alerts while protecting ground stations against hallucinations and rate limits.

Limitation	Description
No cross-pass baseline	Trend detection requires 3 consecutive passes, leaving the first 2 passes without a historical trend baseline.
Synthetic data	Operating ranges are based on representative assumptions rather than a specific operational flight model.

Language model variance	The prose reasoning narrative generated by the model is non-deterministic and can vary in tone.
Heuristic confidence penalties	The score is an explainability metric showing uncertainty sources rather than a statistical probability.
Single ground station model	The link margin does not model elevation angles or orbital geometry.
No inter-subsystem coupling	Subsystems are treated independently. This coupling was simplified due to scope constraints, as modeling full inter-subsystem coupling introduces system identification complexity.
Solar current limit	Near-zero solar current in sunlight is set to a warning floor rather than a critical floor.

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